

Operating System Fundamentals

1. What is an Operating System?

An Operating System is a software that manages and handles hardware and software resources of a computing device.

Function	Description
Resource management	Manages computer resources such as CPU, memory and files.
Interface	Acts as an interface between user and hardware.
Core functions	Performs core functions like processor, memory and file management.
Examples	Linux, Windows, Mac, Android.

2. Introduction to Operating System

An Operating System acts as an intermediary between the computer hardware and the user. It prevents the user from directly accessing the hardware and manages all system resources.

The purpose of OS is to provide an environment in which user can executes programs conveniently and efficiently.

The OS is a program that runs at all times on a computer. All other programs including application programs run on top of the operating system.

It assigns resources such as memory, processors and input/output devices to processes that need them. The assignment of resources has to be fair and secure.

3. Operating System and User Interaction

Every general-purpose computer consists of hardware, an operating system, system programs and application programs.

Part	Examples / Description
Hardware	CPU, ALU, memory, I/O devices, peripheral devices and storage devices.
System programs	Compilers, loaders, editors and the OS itself.
Application programs	User-level software that performs specific tasks.

An operating system is always running in the background to manage and coordinate the use of hardware resources among system and application programs for multiple users.

The OS itself is not only a user interface. Instead, it provides an interface for user interaction, typically through:

- Command-Line Interface (CLI), e.g., Bash and PowerShell.
- Graphical User Interface (GUI), e.g., Windows desktop and macOS Finder.

Kernel: At the core of the OS lies the Kernel, which is the primary interface between hardware and software and handles low-level operations such as process management, memory management, file system control and device handling.
